

Healthcare Workforce Mental Health Analysis Using SQL

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Tools Used:

- Microsoft Excel
- SQL Server Management Studio (SSMS)

Project Objective:

To analyse healthcare workforce mental health trends, stress levels, burnout frequency, job satisfaction, and turnover intention using SQL queries and data analytics techniques.

Question 1

How many healthcare employees are in the dataset?

Query 1;

```
SELECT  
    COUNT(*) AS Total_Healthcare_Workers  
FROM HealthcareMentalHealth;
```

Result;

	Total_Healthcare_Workers
1	5000

Insight:

This analysis helps determine the total number of healthcare workers included in the study, providing a clear understanding of the dataset size used for the mental health assessment.

Question 2

Which healthcare profession has the highest number of employees?

Query2;

```
SELECT  
    Employee_Type,  
    COUNT(*) AS Total_Employees  
FROM HealthcareMentalHealth  
GROUP BY Employee_Type  
ORDER BY Total_Employees DESC;
```

	Employee_Type	Total_Employees
1	Registered Nurse	1283
2	Medical Assistant	513
3	Healthcare Administrator	507
4	Nursing Assistant	502
5	Critical Care Nurse	488
6	Radiology Technician	376
7	Physician (General)	372
8	Laboratory Technician	354
9	Physician (Specialist)	343
10	Pediatric Nurse	262

Insight:

The result highlights the most represented healthcare profession in the organization and helps identify which employee group makes up the largest portion of the workforce.

Question 3

Which department experiences the highest stress level?

Query 3;

```
SELECT  
    Employee_Type,  
    COUNT(*) AS Total_Employees  
FROM HealthcareMentalHealth  
GROUP BY Employee_Type  
  
ORDER BY Total_Employees DESC;
```

Result;

	Department	Average_Stress_Level
1	ICU	8
2	Outpatient Services	7
3	Specialty Care	7
4	General Practice	7
5	Pediatrics	7
6	General Medicine	7
7	Assisted Living	7
8	Radiology	7
9	Laboratory	7
10	Administration	5

Insight:

This analysis reveals the departments where employees are experiencing the greatest levels of stress, which may indicate heavy workload, pressure, or poor working conditions.

Question 4

Which workplace factor contributes most to employee stress?

Query 4;

SELECT

Workplace_Factor,

AVG(Stress_Level) AS Average_Stress_Level

FROM HealthcareMentalHealth

GROUP BY Workplace_Factor

ORDER BY Average_Stress_Level DESC;

Result;

	Workplace_Factor	Average_Stress_Level
1	Heavy Workload	8
2	Work-Life Imbalance	8
3	Safety Concerns	8
4	Poor Work Environment	6
5	Emotional Demands	6
6	Job Insecurity	5
7	Career Stagnation	4
8	Unclear Job Expectations	4

Insight:

The findings help identify the major workplace conditions affecting employee mental well-being and show which organizational factors contribute most to stress among healthcare workers.

Question 5

Which employee type experiences the highest burnout frequency?

Query5;

SELECT

Employee_Type,

COUNT(Burnout_Frequency) **AS** Burnout_Count

FROM HealthcareMentalHealth

GROUP BY Employee_Type

ORDER BY Burnout_Count **DESC**;

Result;

	Employee_Type	Burnout_Count
1	Registered Nurse	1283
2	Medical Assistant	513
3	Healthcare Administrator	507
4	Nursing Assistant	502
5	Critical Care Nurse	488
6	Radiology Technician	376
7	Physician (General)	372
8	Laboratory Technician	354
9	Physician (Specialist)	343
10	Pediatric Nurse	262

Insight:

This result identifies the healthcare professionals most affected by burnout, helping management understand which employee groups may require more support and intervention.

Question 6

What is the average job satisfaction level across departments?

Query 6;

SELECT

Department,

AVG(Job_Satisfaction) AS Average_Job_Satisfaction

FROM HealthcareMentalHealth

GROUP BY Department

ORDER BY Average_Job_Satisfaction DESC;

Result ;

	Department	Average_Job_Satisfaction
1	Pediatrics	2
2	General Medicine	2
3	Assisted Living	2
4	Administration	2
5	Radiology	2
6	Laboratory	2
7	ICU	2
8	Outpatient Services	2
9	Specialty Care	2
10	General Practice	2

Insight:

This analysis evaluates employee satisfaction within different departments and helps determine which areas of the organization have better or lower workplace morale.

Question 7

Which departments record the highest mental health absences?

Query 7;

```
SELECT  
    Department,  
    SUM(Mental_Health_Absences) AS  
    Total_Mental_Health_Absences  
FROM HealthcareMentalHealth  
GROUP BY Department  
ORDER BY Total_Mental_Health_Absences DESC;
```

Result;

	Department	Total_Mental_Health_Absences
1	General Medicine	9900
2	ICU	3985
3	Assisted Living	3848
4	Outpatient Services	3846
5	General Practice	2881
6	Administration	2835
7	Radiology	2634
8	Specialty Care	2594
9	Laboratory	2546
10	Pediatrics	1912

Insight:

The result shows which departments experience the highest number of mental health-related absences, indicating possible stress overload and reduced employee well-being.

Question 8

Does access to Employee Assistance Programs (EAPs) reduce stress levels?

Query 8;

SELECT

Access_to_EAPs,

AVG(Stress_Level) AS Average_Stress_Level

FROM HealthcareMentalHealth

GROUP BY Access_to_EAPs;

Result;

	Access_to_EAPs	Average_Stress_Level
1	0	7
2	1	7

Question 9

Which employee groups have the highest turnover intention?

Query 9;


```

SELECT
    Employee_Type,
    COUNT(*) AS Total_Turnover_Intention
FROM HealthcareMentalHealth
WHERE Turnover_Intention = 1
GROUP BY Employee_Type
ORDER BY Total_Turnover_Intention DESC;

```

Result;

	Employee_Type	Total_Turnover_Intention
1	Registered Nurse	894
2	Medical Assistant	348
3	Critical Care Nurse	347
4	Nursing Assistant	343
5	Physician (General)	278
6	Healthcare Administrator	277
7	Radiology Technician	238
8	Physician (Specialist)	220
9	Laboratory Technician	211
10	Pediatric Nurse	179

Insight;

The findings identify the healthcare professionals most likely to leave their jobs, which may suggest dissatisfaction, burnout, or workplace challenges affecting staff retention.

Question 10

What is the relationship between burnout frequency and stress levels?

Query10;

SELECT

Burnout_Frequency,

AVG(Stress_Level) AS Average_Stress_Level

FROM HealthcareMentalHealth

GROUP BY Burnout_Frequency

ORDER BY Average_Stress_Level DESC;

Result;

	Burnout_Frequency	Average_Stress_Level
1	Often	7
2	Occasionally	7
3	Never	6

Insight:

This analysis shows how increasing burnout frequency may contribute to higher stress levels among healthcare workers, highlighting the connection between emotional exhaustion and workplace pressure.